

Microeconomics

Lecture 16: Monopoly

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What We Know

In previous lectures we analyzed the model of **perfect competition**:

- Many small firms, none with market power; price-taking firm
- At the competitive equilibrium: $P^* = MC$
- First Welfare Theorem: the competitive equilibrium maximizes $TS = CS + PS$
- Any distortion (taxes, administered prices, subsidies) creates $DWL > 0$

In **reality** many markets are not competitive: monopoly (patented drugs, water services), oligopoly (telecommunications, automotive—L18), monopolistic competition (restaurants, clothing).

What we will do

We analyze the extreme case: a firm that is **the only seller** in the market. How does it set the price? How much does it produce? What is the cost for social welfare?

Lecture Objectives

1. **Define** monopoly and identify its sources (barriers to entry)
2. **Explain** why the monopolist faces a downward-sloping demand and $MR < P$
3. **Derive** the marginal revenue formula and its relationship with elasticity
4. **Apply** the $MR = MC$ rule to find the monopoly price and quantity
5. **Compute** consumer surplus, profit, and deadweight loss; compare with PC
6. **Identify** the three structural sources of monopoly power
7. **Explain** why there is no supply curve under monopoly
8. **Analyze** the effect of a specific tax: incidence, revenue, additional DWL
9. **Determine** the optimal allocation of output across several plants ($MR = MC_1 = \dots = MC_n$)
10. **Evaluate** the effects of price regulation and the case of natural monopoly
11. **Solve** complete numerical exercises on basic monopoly, taxation, and multi-plant production

Focus of the lecture

The monopolist is not a *price-taker*: it **chooses the price** (or, equivalently, the quantity). This single difference from the competitive model radically changes the analysis.

What Is a Monopoly?

Monopoly

A **monopoly** is a market structure with a single firm producing a good with no close substitutes, protected by barriers to entry that prevent competition.

Fundamental features: single seller (no direct competitor); *price-maker* (chooses price or, equivalently, quantity); barriers to entry that exclude new entrants; faces a downward-sloping demand.

Examples of monopoly

- **Natural:** water network, electricity distribution, railway tracks (huge fixed costs, increasing returns)
- **Legal:** pharmaceutical patents (Pfizer/COVID-19), copyrights
- **Resource-based:** De Beers in diamonds, OPEC in oil (cartel)
- **Innovation/network-based:** Google search, Microsoft OS (historically)

Barriers to Entry

Why don't new firms enter and erode the monopolist's profit?

Legal barriers:

- Patents (20 years of exclusivity, e.g. drugs)
- Licenses and concessions (frequencies, lotteries)
- Copyright on creative works

Structural barriers:

- Natural monopoly (huge fixed costs)
- Control of an essential input

Strategic barriers:

- Economies of scale
- Sunk costs (unrecoverable investments)
- Reputation and brand loyalty
- Network effects (social media, marketplaces)

Consequence

Without potential competition, the monopolist keeps $\pi > 0$ even in the long run—unlike PC, where free entry drives $\pi \rightarrow 0$.

Sources of Monopoly Power (1/2)

Beyond barriers to entry, a firm's market power depends on **three structural factors**. The first is the most fundamental.

1. The elasticity of market demand

The firm's demand is *at least as elastic* as market demand (in absolute value). Hence $|\varepsilon_D^{\text{market}}|$ sets a **lower bound** on each firm's elasticity, and consequently an **upper bound** on monopoly power in the industry.

Practical implication. If market demand is inelastic, even a single producer (cartel, monopolist) can charge a high markup. If market demand is elastic, nobody—not even a pure monopolist—can charge a significant markup.

OPEC vs. other commodity cartels

OPEC's success. In the 1970s and early 1980s OPEC was able to raise the price of crude oil well above MC because oil demand is inelastic in the short run (few substitutes, capital-intensive).

Failed attempts. Similar cartels in coffee, cocoa, aluminum, and copper failed to sustain high prices: the demand for these goods is much more elastic.

Sources of Monopoly Power (2/2)

2. The number of firms in the market. Other things equal, more firms $\Rightarrow$ less monopoly power for each. When a few firms control most of the sales, the market is said to be **highly concentrated**.

Barriers to entry determine the number of firms in the long run. Typical barriers: patents, copyrights, and licenses (of limited duration); economies of scale (sometimes so large that a single firm is efficient $\rightarrow$ natural monopoly).

3. The interaction among firms. Even with few competitors, the outcome depends on *how* they interact:

- Aggressive price competition $\Rightarrow$ monopoly power evaporates (extreme case: Bertrand $\rightarrow P = MC$ with only 2 firms, L18)
- Collusion (explicit or tacit) $\Rightarrow$ the group replicates the monopoly outcome, with high market power (L18–L19)

Monopoly power is dynamic

Market power *changes over time*: high profits in the SR attract new entrants and tend to erode power in the LR, unless barriers are stable. It is therefore a balance between opposing forces, not a fixed feature.

The Monopolist Faces the Market Demand

The key difference from perfect competition:

Competitive firm

Faces a **horizontal** demand at price P^* .

To sell one more unit: it does not have to lower the price.

$MR = P$ always.

Monopolist

Faces the **market demand** (downward-sloping).

To sell one more unit: it must lower the price **on all units**.

$MR < P$ always (for $Q > 0$).

Numerical intuition. The monopolist sells 10 units at €20. It wants to sell the 11th unit: it must lower the price to €19 *on all 11 units*.

- Gain: revenue from the 11th unit = +19
- Loss: €1 less on each of the first 10 units = -10
- $MR = 19 - 10 = 9 < P = 19$

Marginal Revenue: Formal Derivation

With a downward-sloping inverse demand $P = P(Q)$:

$$TR(Q) = P(Q) \cdot Q \quad \implies \quad MR = \frac{d(TR)}{dQ} = P(Q) + Q \cdot P'(Q)$$

Since $P'(Q) < 0$:

$$MR = P + Q \cdot P'(Q) < P$$

Special case: linear demand $P = a - bQ$:

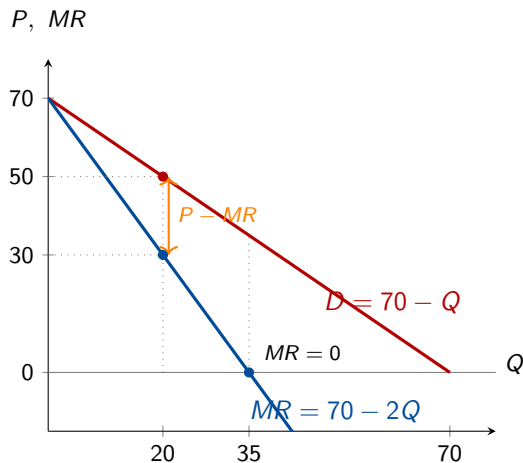
$$TR = aQ - bQ^2 \implies MR = a - 2bQ$$

Twice-the-slope rule (with linear demand)

The MR curve has the **same intercept** as the inverse demand but **twice the slope**. It crosses the quantity axis halfway to where demand does.

Example: if $P = 70 - Q$, then $MR = 70 - 2Q$. At $Q = 35$: $P = 35$, $MR = 0$ (and TR is at its maximum).

MR and Demand: Graph



Properties of MR:

- Same intercept as D (70), twice the slope
- $MR < P$ for every $Q > 0$
- $MR = 0$ at $Q = 35$ (maximum TR)
- $MR < 0$ for $Q > 35$

Check at $Q = 20$, $P = 50$:

The 21st unit sells for 49:

- Direct gain: +49
- Loss of €1 on 20 units: -20
- $MR \approx 49 - 20 = 29 \checkmark$

MR and Elasticity

There is a precise relationship between MR , P , and the price elasticity of demand ε_D :

MR–elasticity formula

$$MR = P \left(1 + \frac{1}{\varepsilon_D} \right) = P \left(1 - \frac{1}{|\varepsilon_D|} \right)$$

The formula follows from $MR = P + QP'(Q)$, recalling that $\varepsilon_D = (dQ/dP) \cdot (P/Q)$.

Immediate consequences:

- $|\varepsilon_D| > 1$ (elastic) $\Rightarrow MR > 0$: increasing Q increases TR
- $|\varepsilon_D| = 1$ (unit elastic) $\Rightarrow MR = 0$: TR is at its maximum
- $|\varepsilon_D| < 1$ (inelastic) $\Rightarrow MR < 0$: increasing Q reduces TR

The monopolist never produces on the inelastic segment

If $MR < 0$, reducing Q increases TR and reduces TC : profit certainly rises. The monopolist moves toward the elastic segment until $MR = MC > 0$.

The $MR = MC$ Rule

The monopolist maximizes profit:

$$\max_{Q \geq 0} \pi(Q) = TR(Q) - TC(Q) = P(Q) \cdot Q - C(Q)$$

First-order condition:

$$\frac{d\pi}{dQ} = MR(Q) - MC(Q) = 0 \implies \boxed{MR(Q_m^*) = MC(Q_m^*)}$$

SOC: $MR'(Q_m^*) < MC'(Q_m^*)$. **Price:** $P_m^* = P(Q_m^*)$ from the inverse demand.

Crucial difference from perfect competition

PC: $P = MC$ (price-taker, $MR = P$). **Monopoly:** $MR = MC$ with $P > MR \Rightarrow P_m^* > MC(Q_m^*)$. The monopolist sets a price *above* marginal cost: this is the source of market power and of the deadweight loss.

Numerical Example: $P = 70 - Q$, $C(Q) = Q^2 + 10Q + 50$

Inverse demand: $P = 70 - Q \Rightarrow MR = 70 - 2Q$.

Marginal cost: $MC(Q) = 2Q + 10$.

FOC: $MR = MC$:

$$70 - 2Q = 2Q + 10 \implies 4Q = 60 \implies Q_m^* = 15$$

Monopoly price:

$$P_m^* = 70 - 15 = 55$$

SOC: $MR' = -2$, $MC' = 2$, and $-2 < 2 \checkmark$ (maximum).

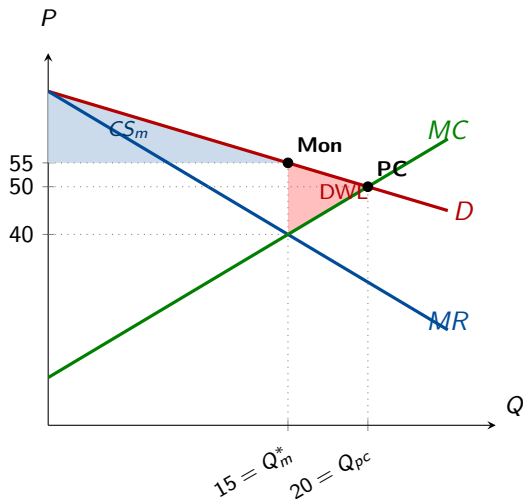
Comparison with perfect competition

Under PC we would have $P = MC$:

$$70 - Q = 2Q + 10 \implies Q_{pc} = 20, \quad P_{pc} = 50$$

The monopoly produces **less** ($15 < 20$) and charges **more** ($55 > 50$) than perfect competition.

Optimization: Graph



Graphical construction:

1. Find Q_m^* from the intersection $MR = MC$
2. Go up *vertically* from Q_m^* to the curve D : there we read $P_m^* = 55$
3. $MC(Q_m^*) = 40 < P_m^* = 55$: positive markup

Key areas:

- CS_m : triangle above P_m^* , below D
- DWL : triangle between D and MC , from Q_m^* to Q_{pc}

Comparison. The monopolist produces less ($15 < 20$) and charges more ($55 > 50$) than under PC.

Welfare under Monopoly

Back to the example: $P = 70 - Q$, $C(Q) = Q^2 + 10Q + 50$, monopoly optimum $Q_m^* = 15$, $P_m^* = 55$.

Consumer surplus:

$$CS_m = \frac{1}{2} Q_m^* (P_{\max} - P_m^*) = \frac{1}{2} \cdot 15 \cdot (70 - 55) = \frac{1}{2} \cdot 15 \cdot 15 = 112.5$$

Monopolist's profit:

$$\pi_m = TR - TC = 55 \cdot 15 - (15^2 + 10 \cdot 15 + 50) = 825 - 425 = 400$$

Check via the integral: $PS_m = \int_0^{15} (P_m^* - MC(Q)) dQ = \int_0^{15} (45 - 2Q) dQ = 675 - 225 = 450$. Hence $\pi_m = PS_m - FC = 450 - 50 = 400 \checkmark$.

Comparison with perfect competition ($Q_{pc} = 20$, $P_{pc} = 50$)

$$CS_{pc} = \frac{1}{2} \cdot 20 \cdot 20 = 200, \quad PS_{pc} = \int_0^{20} (40 - 2Q) dQ = [40Q - Q^2]_0^{20} = 400$$
$$TS_{pc} = CS_{pc} + PS_{pc} = 600$$

Deadweight Loss of Monopoly

Total surplus under monopoly:

$$TS_m = CS_m + PS_m = 112.5 + 450 = 562.5$$

Deadweight loss:

$$DWL = TS_{pc} - TS_m = 600 - 562.5 = 37.5$$

Check with the triangle formula:

$$DWL = \frac{1}{2}(Q_{pc} - Q_m^*)(P_m^* - MC(Q_m^*)) = \frac{1}{2} \cdot 5 \cdot (55 - 40) = \frac{1}{2} \cdot 5 \cdot 15 = 37.5 \checkmark$$

Monopoly redistributes *and* destroys welfare

Compared with PC: $\Delta CS = -87.5$ (consumers lose); $\Delta PS = +50$ (the monopolist gains). The monopolist *captures* 75 of consumer surplus (rectangle $(55 - 50) \cdot 15$) but loses 25 on the units no longer produced: $\Delta CS + \Delta PS = -87.5 + 50 = -37.5$ *vanishes*: this is the DWL.

The DWL of Monopoly: Intuition

Why does monopoly create a deadweight loss?

The monopolist produces 15 units instead of 20. The 5 “missing” units (between $Q_m^* = 15$ and $Q_{pc} = 20$) would have created value for society:

- For each unit between 15 and 20: the value to the consumer (on the demand curve) exceeds the marginal cost
- In particular $P_D(Q) > MC(Q)$ over this whole interval: trades that generate surplus
- The monopolist does not carry them out because increasing Q would require lowering the price *on all units* already sold
- The DWL is exactly the value of the forgone trades that perfect competition would have carried out

In short

Under monopoly, the marginally beneficial trades between Q_m^* and Q_{pc} do not take place. That value goes to nobody (neither consumers, nor the monopolist, nor the government): it is simply lost.

Demand Shifts: No Supply Curve

A counterintuitive property of monopoly. In a competitive market each price P corresponds to a single quantity $Q^S(P)$: the *supply curve*. In a monopoly market this relationship **does not exist**.

Why? The monopolist's choice depends on MR and on MC . When demand shifts, MR changes too, and the combination (P_m^*, Q_m^*) does not follow a fixed path.

Demand shifts can produce four qualitative patterns:

- Same Q , different P (case (a) below)
- Same P , different Q (case (b) below)
- P and Q change in the same direction
- P and Q change in opposite directions

Methodological implication

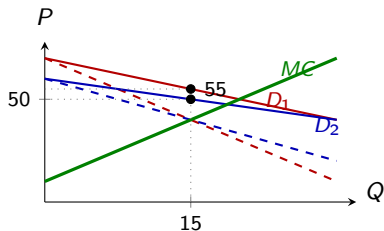
In a monopoly market we cannot reason in terms of “supply meeting demand”. Quantity is determined by $MR = MC$ and the price is read off the demand curve. The same pair (P, Q) can therefore be generated by different demand curves.

Demand Shifts: The Two Typical Cases

Back to $MC = 2Q + 10$ and the initial equilibrium $Q_1^* = 15$, $P_1^* = 55$ with $D_1 : P = 70 - Q$.

Case (a): same Q^* , lower P^*

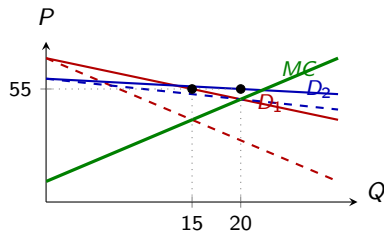
$$D_2 : P = 60 - \frac{2}{3}Q$$



Q unchanged at 15; P falls from 55 to 50.

Case (b): same P^* , larger Q^*

$$D_2 : P = 60 - \frac{1}{4}Q$$



P unchanged at 55; Q rises from 15 to 20 (more elastic demand).

Reading. In both cases the intercept and the slope of demand change. In (a) the new MR_2 crosses MC at the same Q^* , but there the new demand D_2 is lower $\Rightarrow P$ falls. In (b) the new MR_2 crosses MC at a larger Q with more elastic demand $\Rightarrow P$ stays at 55. *Same P , different Q or same Q , different P : no $Q^S(P)$ exists.*

Effect of a Specific Tax on the Monopolist

Setup. The monopolist pays the government a tax t for each unit sold. The effective marginal cost becomes $MC + t$.

New optimality condition:

$$MR(Q) = MC(Q) + t$$

The MC curve shifts upward by t : Q decreases, P increases.

Example. Demand $P = 100 - Q$, $MC = 20$, tax $t = 10$.

- Without the tax: $100 - 2Q = 20 \Rightarrow Q_m^* = 40$, $P_m^* = 60$, $\pi_m = 1,600$
- With the tax: $100 - 2Q = 30 \Rightarrow Q^t = 35$, $P^t = 65$, $\pi^t = 1,225$

Incidence: $\Delta P = 5$ out of $t = 10 \Rightarrow$ consumers bear 50% of the tax; the monopolist the other half.

A surprising result: the monopolist can pass on *more than 100%*

With linear demand, the incidence on consumers is always $\leq 50\%$ ($\Delta P \leq t/2$). But with **convex** demand (e.g. constant elasticity), the monopolist can pass **more than 100%** of the tax on to consumers: $\Delta P > t$. A case that is *impossible* under perfect competition, where $\Delta P \leq t$ always.

Tax on the Monopolist: Revenue and DWL

Continuing the example ($P = 100 - Q$, $MC = 20$, $t = 10$):

Tax revenue: $T = t \cdot Q^t = 10 \cdot 35 = 350$.

Welfare comparison (with PC $\rightarrow Q_{pc} = 80$, $P_{pc} = 20$):

- DWL from monopoly alone: $\frac{1}{2}(Q_{pc} - Q_m^*)(P_m^* - MC) = \frac{1}{2} \cdot 40 \cdot 40 = 800$
- DWL monopoly + tax: $\frac{1}{2}(Q_{pc} - Q^t)(P^t - MC) = \frac{1}{2} \cdot 45 \cdot 45 = 1,012.5$
- Additional DWL from the tax: $1,012.5 - 800 = 212.5$

The policymaker's reading

Taxing a monopolist is *more* distortionary than taxing a competitive market: the tax reduces a quantity that is already below the efficient one, and each unit lost has net value $P - MC > 0$ (here additional DWL of 212.5, versus $\frac{1}{2} \cdot 10 \cdot 10 = 50$ under PC with the same t). Moreover, the tax is not borne *entirely* by the monopolist: part of it (here about $175 = (P^t - P_m^*) \cdot Q^t$ passed on to consumers) falls on them. For a tax under monopoly, incidence depends on the *curvature* of demand, not only on elasticities.

The Multi-Plant Monopolist

Setup. A single monopoly firm owns n plants with different costs $C_1(Q_1), \dots, C_n(Q_n)$. How much should it produce in total, and how should it allocate production?

Two-step principle (intuitive):

1. **Efficient allocation across plants.** For any total Q_T , the firm minimizes total cost:
 $MC_1(Q_1) = MC_2(Q_2) = \dots = MC_n(Q_n)$. Otherwise it could shift a unit from the more expensive plant to the cheaper one.
2. **Optimal total quantity.** The overall MR (based on $Q_T = \sum Q_i$) must equal the common MC .

Optimality condition (multi-plant monopoly firm)

$$MR(Q_T) = MC_1(Q_1) = MC_2(Q_2) = \dots = MC_n(Q_n)$$

Special case. If the plants have *constant* and different marginal costs, the firm uses only the plant with the lowest MC (because the principle requires equal MC s, but with constant MC s this is achieved by using only one).

Multiple Plants: Numerical Example

Data. Market demand $P = 100 - Q$ (hence $MR = 100 - 2Q$). Two plants with costs:

$$C_1(Q_1) = 10 Q_1 + \frac{1}{2} Q_1^2, \quad C_2(Q_2) = 10 Q_2 + Q_2^2$$

$$MC_1 = 10 + Q_1 \quad \text{and} \quad MC_2 = 10 + 2Q_2.$$

Step 1. $MC_1 = MC_2$: $10 + Q_1 = 10 + 2Q_2 \Rightarrow Q_1 = 2Q_2$. Plant 1, with the flatter MC , produces more.

Step 2. $MR(Q_T) = MC_1$, where $Q_T = Q_1 + Q_2 = 3Q_2$:

$$100 - 2(3Q_2) = 10 + 2Q_2 \Rightarrow 90 = 8Q_2 \Rightarrow Q_2^* = 11.25$$

$$Q_1^* = 22.5, \quad Q_T^* = 33.75, \quad P^* = 100 - 33.75 = 66.25$$

Check. $MC_1 = 10 + 22.5 = 32.5$, $MC_2 = 10 + 22.5 = 32.5$, $MR = 100 - 67.5 = 32.5$ ✓

Profit. $TR = 66.25 \cdot 33.75 = 2,235.94$; $C_1 = 478.13$; $C_2 = 239.06$; $\pi = 1,518.75$.

Why Regulate Monopoly?

Monopoly generates DWL, positive profits, and prices above MC . The government can intervene by imposing a **price ceiling** (*price cap*).

Mechanics of the price cap $\bar{P}$. Under regulation, the firm can charge at most $\bar{P}$. The *effective average revenue* curve becomes:

$$AR^{\text{eff}}(Q) = \min \{ \bar{P}, P(Q) \}$$

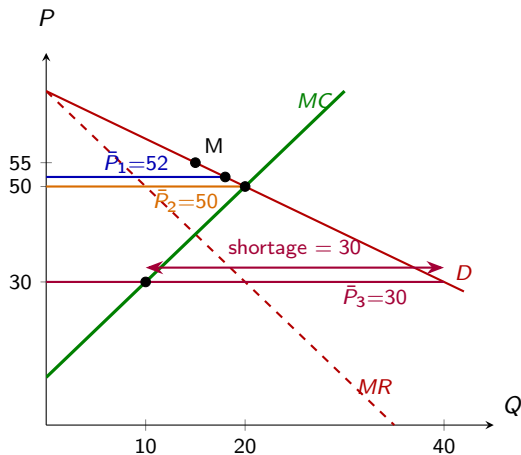
and the *effective marginal revenue* curve is a kinked line:

- For Q such that $P(Q) \geq \bar{P}$ (binding cap): $MR^{\text{eff}} = \bar{P}$ (constant)
- For Q such that $P(Q) < \bar{P}$ (non-binding cap): $MR^{\text{eff}} = \text{original } MR(Q)$

Depending on how low $\bar{P}$ is, we can distinguish **three regimes**:

1. **Intermediate cap** ($P_c < \bar{P} < P_m$): Q rises, P falls. Closer to the competitive outcome.
2. **Competitive cap** ($\bar{P} = P_c$): Q reaches the competitive level Q_c . Social *optimum*.
3. **Cap too low** ($\bar{P} < P_c$): the firm reduces Q below Q_c (and below Q_m if $\bar{P} < MC(Q_m)$), but demand at $\bar{P}$ is even larger $\Rightarrow$ **shortage**.

Price Cap: Numerical Example ($P = 70 - Q$, $MC = 2Q + 10$)



Initial equilibrium (monopoly): $Q_m = 15$, $P_m = 55$.

Case 1 — intermediate cap $\bar{P}_1 = 52$:
 $MR^{\text{eff}} = 52$ up to $\hat{Q}$ such that $P(\hat{Q}) = 52$, i.e. $\hat{Q} = 18$.

$\Rightarrow Q^* = 18$, $P^* = 52$. More output, less DWL.

Case 2 — competitive cap $\bar{P}_2 = 50 = P_c$:
 $\bar{P}_2 = MC(Q_c) = 50. \Rightarrow Q^* = 20 = Q_{pc}$,
 $P^* = 50$. Perfect competition is replicated.
DWL eliminated.

Case 3 — cap too low $\bar{P}_3 = 30$:
 Firm: $\bar{P}_3 = MC \Rightarrow Q^* = 10$ (below Q_m !).
 Consumers: $Q^D(30) = 40$.
 $\Rightarrow$ **shortage of 30 units**; the cap is counterproductive.

Natural Monopoly

Natural monopoly

A **natural monopoly** arises when a single firm can serve the entire market at a *lower* cost than several firms would, thanks to economies of scale extending over the whole output range (high FC and low or decreasing MC).

Examples. Water and sewer networks, electricity and gas distribution, railways, cable TV transmission. Duplicating the infrastructure would be inefficient.

The regulation problem. We would like to impose $P = MC$ (allocative efficiency). But in a natural monopoly AC is **decreasing** over the whole relevant range, so $MC < AC$ always. Imposing $P = MC$ would give $P < AC \Rightarrow$ negative profits: **the firm would go bankrupt**, unless publicly subsidized.

Standard solution: $P = AC$ (*average-cost pricing*)

Set $\bar{P} = AC(Q)$ at the point where the demand curve intersects AC . The firm produces the largest Q at which it stays in business, with **zero economic profit** (it covers all costs, including the normal return on capital). The DWL shrinks but does not vanish: the triangle between Q_r and the efficient allocation $Q^{MC=P}$ remains.

Natural Monopoly: Numerical Example

Data. $P = 25 - Q$, $C(Q) = 64 + 5Q \Rightarrow MC = 5$, $AC(Q) = 64/Q + 5$, $MR = 25 - 2Q$.

(i) Unregulated monopoly:

$$25 - 2Q = 5 \Rightarrow Q_m = 10, P_m = 15.$$

$$\pi_m = (15 - 5) \cdot 10 - 64 = 36.$$

(ii) $P = MC$ (first-best):

$$Q^{FB} = 20, \bar{P} = 5. \text{ But } AC(20) = 8.2 > 5:$$

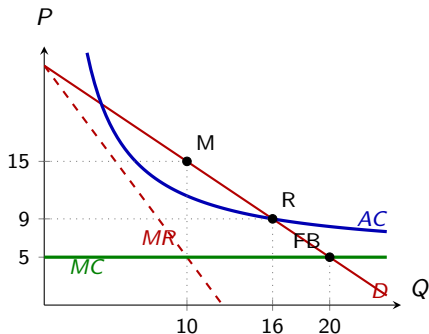
$$\pi = -64 \Rightarrow \text{bankruptcy.}$$

(iii) $P = AC$ (second-best):

$$25 - Q = 64/Q + 5 \Rightarrow Q^2 - 20Q + 64 = 0$$

$$\Rightarrow Q \in \{4, 16\}. \text{ We choose } Q_r = 16.$$

$$P_r = 9, \pi_r = (9 - 5) \cdot 16 - 64 = 0.$$



Assessment. The regulator moves Q from 10 to 16 (+60%), P from 15 to 9 (-40%), excess π to zero. First-best efficiency ($Q = 20$) cannot be achieved without a subsidy: $P = AC$ is the *second-best*.

Exercise 1 — Basic Monopoly

A monopolist faces market demand $P = 80 - Q$ and has the cost function

$$C(Q) = Q^2 + 8Q + 60.$$

- (a) Write the marginal revenue MR and the marginal cost MC .
- (b) Find Q_m^* and P_m^* that maximize profit.
- (c) Compute the profit π_m .
- (d) Compare with the PC optimum: find Q_{pc} , P_{pc} and compute the DWL of monopoly.

Exercise 1 — Solution (a) and (b)

(a) $P = 80 - Q$:

$$TR = P \cdot Q = 80Q - Q^2 \implies MR = 80 - 2Q$$

$$MC = C'(Q) = 2Q + 8$$

(b) FOC: $MR = MC$:

$$80 - 2Q = 2Q + 8 \implies 4Q = 72 \implies Q_m^* = 18$$

$$P_m^* = 80 - 18 = 62$$

SOC check: $MR' = -2$, $MC' = 2$. $MR' < MC'$ ✓ (maximum).

Exercise 1 — Solution (c) and (d)

(c) $C(Q_m^*) = 18^2 + 8 \cdot 18 + 60 = 324 + 144 + 60 = 528.$

$$\pi_m = P_m^* \cdot Q_m^* - C(Q_m^*) = 62 \cdot 18 - 528 = 1,116 - 528 = 588$$

(d) **Perfect competition:** $P = MC \Rightarrow 80 - Q = 2Q + 8 \Rightarrow 72 = 3Q \Rightarrow Q_{pc} = 24, P_{pc} = 56.$

$$MC(Q_m^*) = 2 \cdot 18 + 8 = 44.$$

$$DWL = \frac{1}{2}(Q_{pc} - Q_m^*)(P_m^* - MC(Q_m^*)) = \frac{1}{2} \cdot 6 \cdot 18 = 54$$

Summary. The monopolist produces 18 versus 24 under PC (a 25% reduction), charges a price 10.7% higher (62 vs 56), and creates a deadweight loss of 54.

Exercise 2 — Tax on the Monopolist

Market demand $P = 50 - Q$. Cost function $C(Q) = 4Q$ (hence $MC = 4$, $FC = 0$). The government introduces a specific tax $t = 6$ on each unit sold.

- (a) Find Q_m^* , P_m^* , and π_m *without* the tax.
- (b) Find Q^t , P^t , and π^t *with* the tax.
- (c) Compute the tax revenue T and the incidence of the tax on consumers (share of t passed on to the price).
- (d) Compute the total DWL before and after the tax. How large is the additional efficiency loss attributable to the tax?

Exercise 2 — Solution (a) and (b)

(a) Without the tax. $MR = 50 - 2Q = MC = 4$:

$$Q_m^* = \frac{50 - 4}{2} = 23, \quad P_m^* = 50 - 23 = 27, \quad \pi_m = (27 - 4) \cdot 23 = 529$$

(b) With the tax $t = 6$. The firm maximizes $\pi = (P - MC - t)Q$. New condition $MR = MC + t$:

$$50 - 2Q = 4 + 6 = 10 \implies Q^t = \frac{40}{2} = 20$$

$$P^t = 50 - 20 = 30, \quad \pi^t = (30 - 4 - 6) \cdot 20 = 20 \cdot 20 = 400$$

Effect: quantity falls from 23 to 20 ($\Delta Q = -3$), the price rises from 27 to 30 ($\Delta P = +3$), profit falls from 529 to 400 ($\Delta \pi = -129$).

Exercise 2 — Solution (c) and (d)

(c) **Revenue and incidence.**

$$T = t \cdot Q^t = 6 \cdot 20 = 120$$

$\Delta P = 3$ out of $t = 6$: consumers pay **half** of the tax (50%); the monopolist absorbs the other half. (Standard result with linear demand and constant MC .)

(d) **DWL.** Under PC $P = MC = 4 \Rightarrow Q_{pc} = 46$. Hence:

$$DWL_{\text{monop}} = \frac{1}{2}(46 - 23)(27 - 4) = \frac{1}{2} \cdot 23 \cdot 23 = 264.5$$

$$DWL_{\text{monop}+t} = \frac{1}{2}(46 - 20)(30 - 4) = \frac{1}{2} \cdot 26 \cdot 26 = 338$$

Additional DWL from the tax: $338 - 264.5 = 73.5$.

Reading. The tax on a monopolist raises revenue (120), reduces the monopolist's profit (from 529 to 400), and raises the price for consumers (from 27 to 30). The additional DWL is 73.5, already net of revenue (which is a pure transfer). The distortion from the tax *adds* to the pre-existing DWL of monopoly.

Exercise 3 — Monopoly with Two Plants

A monopolist produces a single good in two separate plants. Market demand is $P = 64 - Q$ (with $Q = Q_1 + Q_2$). The cost functions of the two plants are:

$$C_1(Q_1) = 4 Q_1 + Q_1^2, \quad C_2(Q_2) = 4 Q_2 + 2 Q_2^2$$

- (a) Write MC_1 , MC_2 , and MR .
- (b) Apply the principle $MR = MC_1 = MC_2$ to find Q_1^* , Q_2^* , Q_T^* , and P^* .
- (c) Verify numerically that MC_1 , MC_2 , and MR coincide at the optimum.
- (d) Compute the monopolist's total profit.

Exercise 3 — Solution (a) and (b)

(a)

$$MC_1 = 4 + 2Q_1, \quad MC_2 = 4 + 4Q_2, \quad MR(Q_T) = 64 - 2Q_T$$

(b) *Step 1 (efficient allocation):* $MC_1 = MC_2$:

$$4 + 2Q_1 = 4 + 4Q_2 \implies Q_1 = 2Q_2$$

Step 2 (total quantity): $MR = MC_1$, with $Q_T = Q_1 + Q_2 = 3Q_2$:

$$64 - 2(3Q_2) = 4 + 4Q_2 \implies 60 = 10Q_2 \implies Q_2^* = 6$$

$$Q_1^* = 2 \cdot 6 = 12, \quad Q_T^* = 18, \quad P^* = 64 - 18 = 46$$

Exercise 3 — Solution (c) and (d)

(c) Numerical check.

$$MC_1 = 4 + 2 \cdot 12 = 28, \quad MC_2 = 4 + 4 \cdot 6 = 28, \quad MR = 64 - 2 \cdot 18 = 28$$

$MR = MC_1 = MC_2 = 28$ ✓ (the three marginals coincide at the optimum).

(d) Profit.

$$TR = 46 \cdot 18 = 828$$

$$C_1(12) = 4 \cdot 12 + 12^2 = 48 + 144 = 192$$

$$C_2(6) = 4 \cdot 6 + 2 \cdot 6^2 = 24 + 72 = 96$$

$$\pi = TR - C_1 - C_2 = 828 - 192 - 96 = 540$$

Reading. Plant 1 (with the flatter MC : $\partial MC_1 / \partial Q_1 = 2$ vs $\partial MC_2 / \partial Q_2 = 4$) produces *twice* as much as plant 2. This reflects the rule: the plant with the flatter MC curve takes on a larger share of total output.

General Summary (1/2)

Features of monopoly

- Single seller protected by barriers to entry; price-maker
- Faces the downward-sloping market demand

Marginal revenue

- $MR = P + Q P'(Q) < P$ for every $Q > 0$
- With $P = a - bQ$: $MR = a - 2bQ$ (same intercept, twice the slope)
- $MR = P(1 - 1/|\epsilon_D|)$: the monopolist never produces where $|\epsilon_D| < 1$

General Summary (2/2)

Optimization and welfare

- $MR = MC \Rightarrow Q_m^*$, then $P_m^* = P(Q_m^*) > MC$ (positive markup)
- Compared with PC: lower Q , higher P , $\pi > 0$ in the LR
- $DWL = \frac{1}{2}(Q_{pc} - Q_m^*)(P_m^* - MC(Q_m^*))$: monopoly redistributes and destroys welfare

Sources of monopoly power

- Elasticity of market demand (upper bound on each firm's power)
- Number of firms and market concentration
- Mode of interaction (competition vs collusion)

Further topics and regulation

- No supply curve under monopoly: same Q at different P , or same P at different Q
- Specific tax t : $MR = MC + t$; incidence depends on the curvature of demand ($\Delta P > t$ possible)
- Multiple plants: $MR(Q_T) = MC_1 = \dots = MC_n$ (equalize MC s, then common $MR = MC$)
- Price cap: three possible regimes (closer to PC, replicating PC, shortage if too low)
- Natural monopoly: $P = MC$ leads to bankruptcy; second-best solution $P = AC$ with $\pi = 0$

Skills Acquired

- ✓ Recognize the features of a monopoly market and classify barriers to entry
- ✓ Identify the three structural sources of monopoly power (market elasticity, number of firms, interaction)
- ✓ Derive the marginal revenue curve and the relationship $MR = P(1 - 1/|\varepsilon_D|)$
- ✓ Apply the $MR = MC$ rule; compute CS , π , DWL and compare with PC
- ✓ Explain why there is no supply curve under monopoly
- ✓ Analyze the effect of a specific tax: incidence, revenue, and additional DWL
- ✓ Determine the optimal allocation across several plants via $MR = MC_1 = \dots = MC_n$
- ✓ Distinguish the three price-cap regimes and identify the natural monopoly case ($P = AC$ second-best)
- ✓ Solve exercises on basic monopoly, taxing the monopolist, and multi-plant production

Optional Homework Exercises

1. Monopolist with $P = 96 - 2Q$ and $C(Q) = Q^2 + 12Q + 30$. Find Q_m^* , P_m^* , π_m . Compare with perfect competition and compute the DWL.
2. Monopolist with constant $MC = 5$ and inverse demand $P = 50 - Q/3$. Find Q_m^* , P_m^* . Compute $|\varepsilon_D|$ at the optimum. Verify the formula $MR = P(1 - 1/|\varepsilon_D|)$ directly.
3. Prove formally, using the formula $MR = P(1 - 1/|\varepsilon_D|)$, that the monopolist never produces on the inelastic segment of demand.
4. A pharmaceutical company holds a patent on a drug with demand $Q = 100 - P$ and $MC = 10$. Find the monopoly price. How large is the DWL? What maximum price $P_{\max}$ would make Q_m^* coincide with Q_{pc} while keeping $\pi \geq 0$?
5. True or false: “under monopoly, at the optimum we always have $|\varepsilon_D| > 1$ ”. Justify with reference to the relationship between MR , MC , and elasticity.